Data Science Track — Catch-Up Note (Week 6)

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Why this exists

The AnalystLab Africa Week 6 brief requires each intern to complete at least one meaningful cross-track integration activity, and lists **Data Analytics → Data Science** ("analytical findings inform modelling decisions") as one of the expected integration paths.

By the start of Week 6, the assigned Data Science track teammate(s) had not produced any Week 4 or Week 5 output, and were not responsive to outreach. Rather than skip the integration requirement, or fabricate a collaboration that did not happen, I built the Data Science track's work independently — starting from where it should have started in Week 4 (a machine learning problem definition), through where it should have been at the end of Week 5 (a baseline classification model with its own data exploration and validation of the modelling approach), and carrying it through the Week 6 objective (error analysis, feature refinement, an improved model, and validation).

An initial pass at this catch-up work skipped the Week 4 deliverable entirely and went straight to a Week 5 baseline model without its own visual data exploration, leakage check, or train/test-strategy justification. That gap was identified on review and has since been filled — see `Week4_ML_Problem_Definition.md` and `Week5_Baseline_Modelling_Report.md`.

This is documented honestly as independently-produced catch-up work, not as output from a real Data Science teammate. It uses my own Week 4/5 Data Analytics findings as the modelling input, which is the actual integration point — see `Model_Improvement_Validation.md` for how each DA finding maps to a modelling decision.

Scope built

Week 4 DS expectation (not delivered by anyone)	Built here
ML problem definition, target, features, cancellation handling, initial approach	`scripts/prep_features.py` (implementation) + `reports/Week4_ML_Problem_Definition.md`
Week 5 DS expectation (not delivered by anyone)	Built here
Data preparation (missing values, duplicates, dtypes, categoricals, irrelevant vars)	`reports/Week5_Baseline_Modelling_Report.md` Section 2

Week 4 DS expectation (not delivered by anyone)	Built here
Data-leakage check	`reports/Week5_Baseline_Modelling_Report.md` Section 3
Visual exploration (histograms, box plots, count plots, correlation heatmap, feature-target plots) with interpretation tied to decisions	`scripts/eda_visual_evidence.py`, `outputs/figures/eda/`, interpreted in `reports/Week5_Baseline_Modelling_Report.md` Section 4
Feature engineering (what/why/expected value)	`reports/Week5_Baseline_Modelling_Report.md` Section 5
Train/test strategy with justification	`reports/Week5_Baseline_Modelling_Report.md` Section 6
Baseline classification model	`scripts/train_baseline.py` — Logistic Regression, `class_weight="balanced"`
Initial evaluation	`outputs/models/baseline_metrics.json`, confusion matrix, ROC curve
Week 6 DS expectation	Built here
Error analysis	`scripts/error_analysis.py` — false positive/negative segment profiling, error rate by lead-time/distance/prior-no-show band
Feature refinement	Double-risk interaction flag, lead-time × prior-no-shows product term, distance-cliff flag (all derived from DA findings, and from the Week 5 correlation-heatmap/box-plot evidence)
Improved model	`scripts/train_improved.py` — Random Forest + a refined Logistic Regression, compared against baseline
Model comparison & evaluation	`outputs/models/model_comparison.csv`, ROC/metric comparison charts
Candidate model recommendation	`outputs/models/candidate_model.json`
Documentation	`reports/Model_Improvement_Validation.md`

What is still missing (real Data Science track dependency)

This catch-up now covers the DS track's full Week 4 → 5 → 6 path as specified in the assignment briefs, built using standard classification technique. What it does **not** replace is independent technical review: no second Data Science practitioner has checked these modelling decisions, and none of the Week 7 validation steps listed in `Model_Improvement_Validation.md` Section 11 (cross-validation,

hyperparameter search, probability calibration) have been run yet. If the real Data Science track teammate becomes available, this should be treated as a starting point for review and revision, not a final answer.